AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT395

Applied Cyber Security

Final Examination
Fall Semester 2025/2026

Date: 15/Jan./2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 h

nstructions:

- Total Marks: 50
- Total Number of Questions:5
- Marks Distribution:
  - Question 1: 10
  - Question 2: 10
  - Question 3: 10
  - Question 4: 10
  - Question 5: 10
- Write your answer in the answer booklets. Answers given on the exam paper w not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.

CamScanner

Question 1: Answer the following questions [10 Marks]

1. Which security system detects intrusions by comparing abnormal behavior with normal system activity?
   a) Host Intrusion Detection System (HIDS)
   b) Packet filter
   c) Internet Relay Chat (IRC)
   d) Personal software firewall
2. Which type of software secretly monitors a user's online activities, collects information without consent, and sends it to a third party?
   a) Malware
   b) Spyware
   c) Adware
   d) All of the above
3. Which wireless security standard offers better protection compared to WEP?
   a) WPA
   b) WEP2
   c) TKIP
   d) MAC filtering
4. In a one-way password file, how is the password stored?
   a) By jumbling the password before storing it.
   b) By XORing the password with a key.
   c) By storing the hash value of the password.
   d) By passing the password through a PRF and storing it.
5. Ian has been offered a role that mainly focuses on determining user security requirements. Which security position best fits this role?
   a) Security administrator
   b) Security technician
   c) Security officer
   d) Security manager
6. A _____ can be a hardware device or a software program that filters all th packets of data that comes through a network, the internet, etc.
   a) Firewall
   b) Antivirus
   c) Malware
   d) Cookies
7. Which of these is a technique that is used to verify a message's integrity?
   a) Message Digest
   b) Protocol
   c) Decryption algorithm

CamScanner

d) Digital signature

8. Before encryption, what type of text is given as input to a cipher algorithm?
   a) Scalar text
   b) Complex text
   c) Transformed text
   d) Plain text
9. Which of the following malware types does not clone or replicate itself through infection?
   a) Viruses
   b) Worms
   c) Trojans
   d) Rootkits
10. Which of these is usually referred to as the port number (default) of several web servers like a pache?
   a) 40
   b) 20
   c) 87
   d) 80

Question 2 Match the cybersecurity term with its corresponding definition: [10 Marks]

1. A hacking technique that attempts to gain access by trying every possible password or key combination.	a) Social engineering
2. Software created to detect, block, and remove malicious programs from computer systems.	b) Encryption
3. A process that protects information by converting it into an unreadable format without a decryption key.	c) Firewall
4. A policy that allows staff to use personal devices while applying organizational security controls.	d) Phishing
5. A protective system that filters and controls incoming and outgoing network traffic.	e) Vulnerability scanning
6. A manipulation-based attack that exploits human behavior to compromise security.	f) Brute force attack
7. A formal procedure used to detect, manage, and recover from cybersecurity incidents.	g) Penetration testing
8. An automated method for identifying known security weaknesses in systems or networks.	h) Antivirus software
9. A network design method that divides a network into smaller isolated segments to reduce risk.	i) Network segmentation
10. A controlled security assessment where attacks are simulated to evaluate system defenses.	j) BYOD (Bring Your Own Device) k) Security incident response

Question 3: Answer the following questions [10 Marks]

CamScanner

1. Your organization is implementing a bring-your-own-device (BYOD) policy. How can employees ensure cybersecurity when using personal devices for work? (4 Marks)
2. If you receive an email claiming that you have won a lottery prize, what actions should you take to protect yourself from potential cyber threats? (4 Marks)
3. How does vulnerability assessment differ from penetration testing?(2Marks)

Question 4: Answer the following questions [10 Marks]

1. How does a Man-in-the-Browser (MITB) attack differ from a MITM attack? (2 Marks)
2. What is the principle of least privilege in access control, and why is it important? (2 Marks)
3. How do SQL injection attacks work, and what are some examples of their impact? (2 Marks)
4. What is a denial of service (DoS) attack, and how does it differ from a distributed denial of service (DDoS) attack? (2 Marks)
5. What is ransomware, and how does it affect a user's device? (2 Marks)

Question 5: Answer the following questions [10 Marks]

1. List a four best practices for managing passwords to enhance security?(2 Marks)
2. What are the main types of malware, and how do they differ? 2 Marks)
3. What is the role of redundancy in achieving fault tolerance? (2 Marks)
4. What are the three essential elements of Business Continuity Planning (BCP)? (2 Marks)
5. What is the purpose of a Business Impact Analysis (BIA)? (2 Marks)

End of Questions

CamScanner